
Continuous-Time Fourier Transform (CTFT)

used for: finite-energy, continuous-time signals ($x(t) \rightarrow X(f)$)

analysis formula: $X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$

synthesis formula: $x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$

symmetries: $x(-t) \xleftrightarrow{\text{CTFT}} X(-f)$

$$x^*(t) \xleftrightarrow{\text{CTFT}} X^*(-f)$$

shifts: $x(t - t_0) \xleftrightarrow{\text{CTFT}} e^{-j2\pi f t_0} X(f)$

$$e^{j2\pi f_0 t} x(t) \xleftrightarrow{\text{CTFT}} X(f - f_0)$$

scaling: $x(at) \xleftrightarrow{\text{CTFT}} \frac{1}{|a|} X\left(\frac{f}{a}\right)$

convolution: $(x * y)(t) \xleftrightarrow{\text{CTFT}} X(f) Y(f)$

modulation: $x(t) y(t) \xleftrightarrow{\text{CTFT}} \int_{-\infty}^{\infty} X(v) Y(f - v) dv$

differentiation: $\frac{d}{dt} x(t) \xleftrightarrow{\text{CTFT}} j2\pi f X(f)$

$$t x(t) \xleftrightarrow{\text{CTFT}} \frac{j}{2\pi} \frac{d}{df} X(f)$$

Parseval's Relation: $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$

Notable CTFT Pairs $(t, f \in \mathbb{R})$

$$x(t) = \delta(t - t_0)$$

$$X(f) = e^{-j2\pi f t_0}$$

$$x(t) = 1$$

$$X(f) = \delta(f)$$

$$x(t) = e^{j2\pi f_0 t}$$

$$X(f) = \delta(f - f_0)$$

$$x(t) = \cos(2\pi f_0 t + \phi)$$

$$X(f) = \frac{1}{2} \left[e^{j\phi} \delta(f - f_0) + e^{-j\phi} \delta(f + f_0) \right]$$

$$x(t) = \sin(2\pi f_0 t + \phi)$$

$$X(f) = \frac{-j}{2} \left[e^{j\phi} \delta(f - f_0) - e^{-j\phi} \delta(f + f_0) \right]$$

$$x(t) = u(t)$$

$$X(f) = \frac{1}{j2\pi f} + \frac{1}{2} \delta(f)$$

$$x(t) = e^{-at} u(t), \quad a > 0$$

$$X(f) = \frac{1}{a + j2\pi f}$$

$$x(t) = \begin{cases} 1, & |t| < T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$X(f) = T \operatorname{sinc}(Tf)$$

$$x(t) = 2f_c \operatorname{sinc}(2f_c t)$$

$$X(f) = \operatorname{rect}\left(\frac{f}{2f_c}\right)$$

$$\text{where } \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

$$\text{and } \operatorname{rect}(x) = \begin{cases} 1, & |x| < \frac{1}{2} \\ 0, & \text{otherwise.} \end{cases}$$